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Group: CRLWND

Assigned Role: Memory Management

What did I contribute to the project? I was responsible for the memory management module of MiniOS. This module controls how the operating system allocates, tracks, and frees memory for the kernel.

What features did I personally develop? I created the MemoryManager class, which keeps track of the system's memory, and connected it to the MiniOS shell by creating a MemoryManager instance (self.memory) and registering three commands: memory, allocate, and free. This lets a user view and control simulated memory from the command line.

What problems did I encounter? Preventing the system from freeing more memory than was allocated, which would make the numbers wrong and connecting my module to the shell so the commands call the right methods.

How did I solve those problems? I used register_command with a usage description (allocate <MB>, free <MB>), so each command is easy to call and appears in the help output. Tested by running allocate and free repeatedly and checking the result with memory.

What did I learn from this project? The basic role of memory management in an operating system: tracking, allocating, and releasing memory and how a shell works, where commands are registered and mapped to handler functions.

Evidence of my contribution:

